Java Output Practice

Topic: for / while / method

Instructions

Read the code and write the **exact output**.

注意：

- 注意 `print` 和 `println` 的区别
- 注意 method 里“打印了什么”和“返回了什么”是两回事
- 注意循环变量每次更新后的值
- 注意有些题的输出会连在同一行

Q1

```java
public class Q14 {
    public static int step(int x) {
        x = x + 3;
        System.out.print(x + " ");
        return x - 1;
    }

    public static void main(String[] args) {
        int a = 2;
        a = step(a);
        System.out.print(a + " ");
        a = step(a);
        System.out.println(a);
    }
}
```

Q2

```java
public class Q15 {
    public static void main(String[] args) {
        for (int i = 1; i <= 3; i++) {
            if (i == 2) {
                System.out.println("middle");
            } else {
                System.out.println(i);
            }
        }
```

```
    }
}
```

Q3

```
public class Q16 {
    public static int m(int x) {
        System.out.print(x);
        return x + 2;
    }

    public static void main(String[] args) {
        int n = 1;

        for (int i = 0; i < 2; i++) {
            n = m(n);
        }

        System.out.println(n);
    }
}
```

Q4

```
public class Q17 {
    public static void main(String[] args) {
        int i = 1;
        int sum = 1;

        while (i <= 3) {
            sum = sum * 2;
            System.out.println(sum);
            i++;
        }
    }
}
```

Q5

```
public class Q18 {
    public static void printNum(int n) {
        System.out.print("[" + n + "]");
    }
```

```
    public static void main(String[] args) {
        for (int i = 2; i <= 4; i++) {
            printNum(i);
        }
        System.out.println();
    }
}
```

Q6

```
public class Q19 {
    public static int calc(int x) {
        if (x % 2 == 0) {
            return x / 2;
        }
        return x + 1;
    }

    public static void main(String[] args) {
        int n = 3;

        for (int i = 0; i < 3; i++) {
            n = calc(n);
            System.out.println(n);
        }
    }
}
```

Q7

```
public class Q20 {
    public static void word(String s, int n) {
        for (int i = 0; i < n; i++) {
            System.out.print(s);
        }
        System.out.println();
    }

    public static void main(String[] args) {
        word("A", 2);
        word("B", 3);
    }
}
```

Q8

```
public class Q21 {
    public static int g(int x) {
        System.out.println("g:" + x);
        return x + 1;
    }

    public static void main(String[] args) {
        int a = 1;
        int b = g(a);
        int c = g(b);
        System.out.println(a + " " + b + " " + c);
    }
}
```

Q9

```
public class Q22 {
    public static void main(String[] args) {
        for (int i = 1; i <= 3; i++) {
            System.out.print(i);

            for (int j = 1; j <= 2; j++) {
                System.out.print(j);
            }

            System.out.println();
        }
    }
}
```

Q10

```
public class Q23 {
    public static int next(int x) {
        System.out.println("in method: " + x);
        x++;
        return x;
    }

    public static void main(String[] args) {
        int n = 0;

        while (n < 3) {
```

```
            n = next(n);
            System.out.println("in main: " + n);
        }
    }
}
```

Q11

```
public class Q24 {
    public static void main(String[] args) {
        int x = 2;

        for (int i = 0; i < 3; i++) {
            x = x + i;
            System.out.println(x);
        }
    }
}
```

Q12

```
public class Q25 {
    public static void box(int n) {
        System.out.print("<");
        for (int i = 0; i < n; i++) {
            System.out.print("-");
        }
        System.out.println(">");
    }

    public static void main(String[] args) {
        box(1);
        box(3);
        box(2);
    }
}
```

Java Output Practice — Method Calls Method

Topic: method calling method / return / print / execution order

Instructions

Read the code and write the **exact output**.

Q13

```java
public class Q26 {
    public static void bar() {
        System.out.println("bar");
    }

    public static void foo() {
        System.out.println("foo start");
        bar();
        System.out.println("foo end");
    }

    public static void main(String[] args) {
        foo();
    }
}
```

Q14

```java
public class Q27 {
    public static int bar(int x) {
        System.out.println("bar: " + x);
        return x + 1;
    }

    public static int foo(int y) {
        System.out.println("foo: " + y);
        return bar(y * 2);
    }

    public static void main(String[] args) {
        int n = foo(3);
        System.out.println("main: " + n);
    }
}
```

Q15

```java
public class Q28 {
    public static void bar(int x) {
        System.out.println("B" + x);
    }
```

```
    public static void foo(int y) {
        System.out.println("F" + y);
        bar(y + 1);
    }

    public static void main(String[] args) {
        foo(2);
        bar(5);
    }
}
```

Q16

```
public class Q29 {
    public static int bar(int x) {
        System.out.print(x);
        return x + 2;
    }

    public static int foo(int y) {
        System.out.print(y);
        return bar(y + 1);
    }

    public static void main(String[] args) {
        int a = foo(1);
        System.out.println(a);
    }
}
```

Q17

```
public class Q30 {
    public static int bar(int x) {
        return x * 2;
    }

    public static int foo(int y) {
        return bar(y) + 1;
    }

    public static void main(String[] args) {
        System.out.println(foo(3));
    }
}
```

Q18

```
public class Q31 {
    public static void bar() {
        System.out.println("inside bar");
    }

    public static void foo() {
        System.out.println("inside foo");
        bar();
    }

    public static void main(String[] args) {
        System.out.println("start");
        foo();
        System.out.println("end");
    }
}
```

Q19

```
public class Q32 {
    public static int bar(int x) {
        System.out.println("bar gets " + x);
        return x - 1;
    }

    public static int foo(int y) {
        System.out.println("foo gets " + y);
        int z = bar(y + 2);
        return z * 2;
    }

    public static void main(String[] args) {
        System.out.println(foo(4));
    }
}
```

Q20

```
public class Q33 {
    public static void bar(int x) {
        System.out.print(x + " ");
    }
```

```
    public static void foo(int y) {
        bar(y);
        bar(y + 1);
    }

    public static void main(String[] args) {
        foo(3);
        System.out.println();
    }
}
```

Java Output Practice — Mixed Review

Topic: if / else + for / while + method

Instructions

Read the code and write the **exact output**.

Q1

```
public class Q1 {
    public static int foo(int x) {
        if (x % 2 == 0) {
            System.out.print("E");
            return x / 2;
        } else {
            System.out.print("O");
            return x + 1;
        }
    }

    public static void main(String[] args) {
        int n = 3;

        for (int i = 0; i < 3; i++) {
            n = foo(n);
            System.out.print(n + " ");
        }

        System.out.println();
    }
}
```

Q2

```java
public class Q2 {
    public static void bar(int x) {
        if (x > 3) {
            System.out.println("big");
        } else {
            System.out.println("small");
        }
    }

    public static int foo(int y) {
        int count = 0;

        while (y > 0) {
            count++;
            y = y - 2;
        }

        return count;
    }

    public static void main(String[] args) {
        int a = foo(5);
        bar(a);

        int b = foo(8);
        bar(b);
    }
}
```

Q3

```java
public class Q3 {
    public static int step(int x) {
        if (x < 5) {
            System.out.print("A");
            return x + 2;
        } else {
            System.out.print("B");
            return x - 3;
        }
    }

    public static void show(int n) {
        while (n > 0) {
            if (n % 2 == 0) {
                System.out.print(n);
            } else {
                System.out.print("*");
            }
```

```
            n--;
        }
        System.out.println();
    }

    public static void main(String[] args) {
        int x = 2;

        for (int i = 0; i < 2; i++) {
            x = step(x);
        }

        System.out.println(x);
        show(x);
    }
}
```